

PATENTED
BooM-O₂
...an water doctor

Co-treatment Water Conditioner



Manufactured by :



Marketed by :



- 
- ♦ **WATER** stands out to be the most essential element in our lives as the animal body constitutes 60% of water
 - ♦ Water is **MOST** essential **NUTRIENT** & Least **FOCUSED**
 - ♦ However, 80% of the global diseases are water borne which arise out of consumption of unsafe water, inadequate sanitation and poor hygiene practice
 - ♦ 2.2 billion people use without safely managed water (WHO)

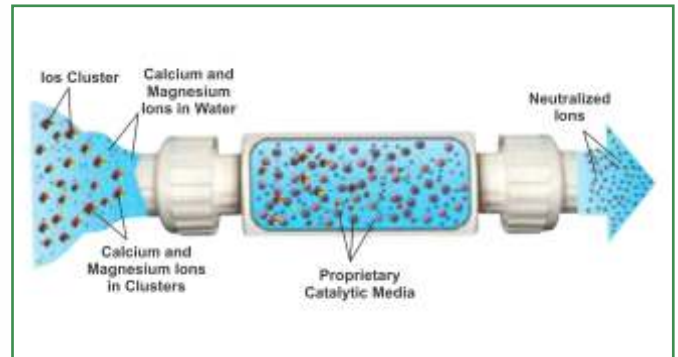
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Co-treatment Water Conditioner

Introducing

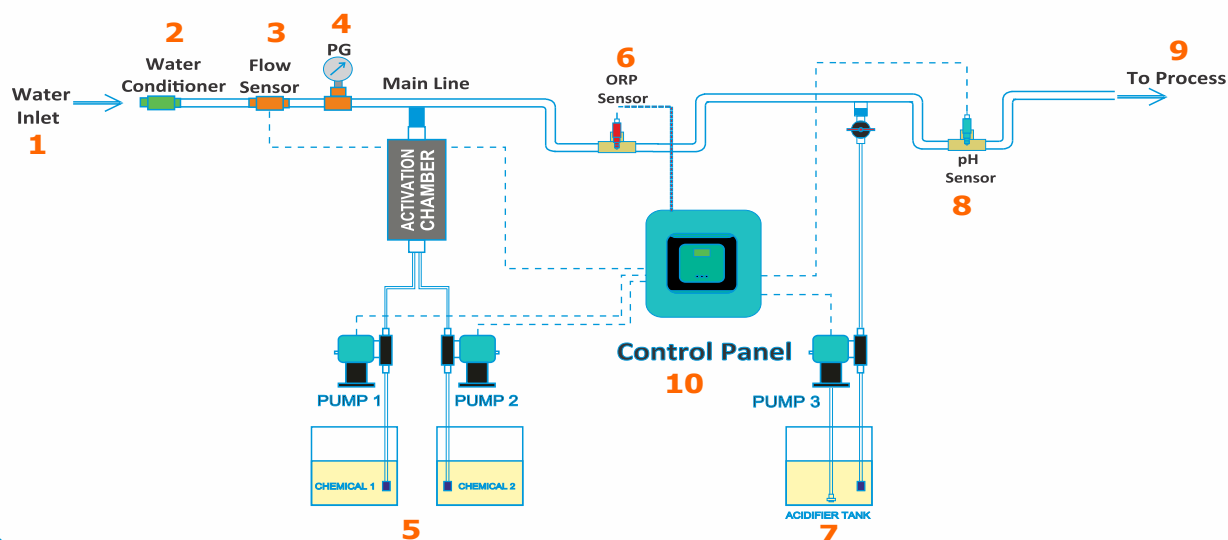


Israeli Technology Based
Water Improver



Description :

- Patented and robust innovative water treatment machine for Hardness, ORP (ClO₂) & pH for 24*7 action.
- Engineering marvel for water treatment !
- It replace all existing solutions available for water treatment & protect livestock from all water born challenges which are major in poultry. It support to reduce economic impact due to poor quality water.



Boom Ox Modus Operandi :

1	Water Inlet	At this point we have to install the machine. Common inlet pipe from tank/source to whole farm.
2	Water Conditioner	This will helps to neutralise majority ions, metals and hardness causing agents.
3	Flow Sensor	Helps to check flow of water and it helps for accurate dosing of medicines.
4	Pressure Gauge	It helps to maintain water pressure.
5	Chlorine DiOxide solutions	These two solutions need to fill & refill once receive signal from device. This solution can't be change by other source, as it is calibrated with brain/control panel of machine for automated doses system
6	ORP Sensor	It helps to maintain desired ORP (650+ mV) as we set in device
7	pH solutions	This acidifier solutions need to fill & refill once receive signal from device. This solution can't be change by other source, as it is calibrated with brain/control panel of machine for automated doses system
8	pH Sensor	It helps to maintain desired pH (5.8-6.0) as we set in device or as needed
9	IoT Device	This device is indicator of all parameters which we set and it send notification messages. - It must be implemented at last shed and last row and last point. So that, water treatment can be done as per set limits of ORP and pH from last point to start point.
10	Control Panel/Brain	This is actual brain of machine. It receive all signals and give solutions as per given commands. Most important and patented instrument.
11	Best ROI	Cost of medicines is almost equal to existing price but it gives much better ROI than traditional ways of water treatment.
12	Entire machine is well assembled, dust proof, water resistance, rodents resistance and durable etc. Size of machine is approx one office table and can mount on wall easily- Maintenance free machine except minor calibration of sensors once in quarter which user can do at its own	
13	User can replaces all of his available solutions for water treatment including pipeline flushing and cleaning	
14	During vaccination and medication, user can stop sensors and use normal non medicated water from same machine	
15	For 1st installation and till smooth operations, company engineer will help	
16	Other much more benefits which keep livestock healthy and owner happy	

Technical Specifications

Features

- Loaded Fuzzy Logic with Patent Pending Technology
- Accurate ORP Controlled Dosing with Level Interlock
- Specially Designed Activator Chamber for Stabilized Chlorine Dioxide in CPVC
- Auto Voltage Regulator
- 3.5" LCD display
- IoT Enabled Process Monitoring

Scope of Supply

- Metering Pump - 3 Nos
- IP 65 Control Panel (Make-Germany)
- ORP & pH Sensor with holder
- Wall Mounting Plate with sensor fittings

Why Boom-Ox

- Very Stable & sustain upto 24 Hrs +
- Highly Soluble Like Despines
- pH Independent
- Lowest CT Value and Fast Acting
- Act as Sterilant (Kills all Organisms Like Bacteria, Viruses, Algae as well as Fungi)

How it Works?

- Boom-Ox Penetrates the cell wall and destructs RNA of micro organism
- Causes Oxidation of Amino Micro Organism
- Boom-Ox Kills Micro organisms even when they are inactive
- Removes Biofilm
- Causes oxidation of micro-organism amino acids leading to it's destruction.

Dosing Pump

- Electrical : 230 V AC {+/-10% } 50HZ
- Suction Tubing : 4/6 mm Discharge
- Tubing : 4/6 mm
- Dosing Rate : 3 LPH @ 4 Kg/cm2
- Wetted Parts : PP
- Ingress Protection : IP65

ORP & pH Sensor

- ORP Range : -1000 to +1000 mV
- pH Range : 0-14
- Body Material : ABS
- Max.Operating Pressure: 5 Kg/cm2
- Cable length : 5 meters
- Mounting Thread : 3/4"

CT VALUES

ORGANISM	ppm of BOOM-OX	Contact Time in Sec.	Killing %
S.aureus	0.5	60	99.99
E.coli	0.25	100	99.99
Streptococcus	0.5	60	99.99
Lactobacillus	0.5	60	99.99
Pseudomonas	1	150	99.99

SPECTRUM OF BOOM-OX AGAINST ORGANISMS

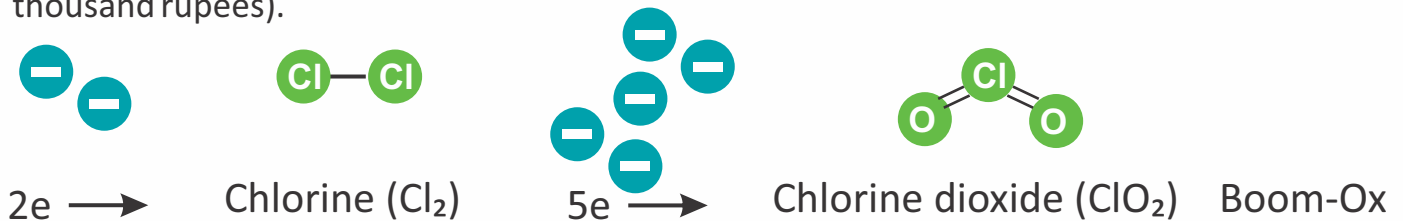
GRAM POSITIVE	GRAM NEGATIVE	PROTOZOA
Clostridium	Escherichia Coli	Cryptosporidium
Staphylococcus	P. multocida	Microsporidium
Streptococcus	S.gallinarum	Giardia

Uniqueness of Boom-ox

- Boom-Ox CLO2 can replace water disinfectant, sanitizer and other solution after discussion with our expert.
- A patented combination of Water Disinfectant and sanitizer
- Maintain ORP & pH continuously : 24 *7.
- All in one patented solution for water treatment, we call it " **Water Doctor**".
- It kills 99.99% water born infections and insuring 100% safe and potable water for livestock. Best in class and economical.
- Boom-Ox covers most of the physical, chemical and biological treatment aspects of water.

Chlorine vs Boom-Ox (Chlorine Dioxide)

The effectiveness of chlorine is dependant on water quality & ph of water. Chlorine tends to bind itself to any organic material in the water, reducing its potency. Chlorine is also less effective if water pH is higher than 7.0 and totally ineffective above pH of 7.4 resulting into water borne diseases (social impact) & money down the drain (financial losses in multi thousand rupees).



CHLORINE	Boom-Ox (Chlorine Dioxide ClO_2)
Slower E-Coli distraction	Faster E-coli distraction due to 5e-(electron) exchange from ClO_2 .
Chlorine is resistant 30 plus pathogens like e-coli, hepatitis A, salmonella bifolia bacteria etc.	There is no resistant to any types of pathogens and bacteria.
Chlorine is ineffective against complex organism such as cysts and protozoa.	Boom-Ox is extreemly effective against complex organisms such as cysts and protozoa.
Chlorine not works in broader pH range.	Boom-Ox is highly effective with high organic matter.
Chlorine has a higher oxidation potential & does the hydrolyse to form an acid and therefore is high corrosive.	Boom-Ox works in broader pH range 4-10.
Not removed the biofilm.	Boom-Ox has lower oxidation potential and does not hydrolyse to form an acid and therefore is less corrosive.
Chlorine is already banned in certain parts of Europe and USA due to its hazards characteristics. India has also restricted the use of Chlorine since 2010, due to the threats it poses to human health.	Biofilm studies & various tests carried out at CAMR and Proton Down have proved that Boom-Ox does it more faster than others solution.
Chlorine has a tendency of reacting with organics in water to form harmful byproducts which are often carciogenic - the consumption of which pose the threat of cancer forms THMs (tri halo Methane) and HAAs (Halo acetic acid) which are considered to be potential human carcinogens.	World wide approved product. Boom-Ox remove bad odour of water.
Chlorine cannot be used at temperature $>40^\circ\text{C}$ as it tends to disassociate evolving Cl_2 gas.	Boom-Ox used at $>40^\circ\text{C}$ as it does not disassociated at eleveted temperature.
	Boom Ox as a most wide spectrum disinfectant will cover entire biological treatment and very small physical and chemical treatment of water by oxidizing Iron, Manganese and Magnesium.

PPM vs ORP (Oxidation Reduction Potential)

- ◆ ORP- stands for Oxidation Reduction Potential. ORP is a measurement, in millivolts (mV), of the water's potential for oxidizers to steal electrons from another substance.
- ◆ Oxidation is the loss of electrons by an atom, molecule or ion. (Often, the lost electrons are replaced by oxygen.)
- ◆ Reduction is the net gain of electrons by an atom, molecule or ion.
- ◆ The rate of electron transfer is measured in millivolts as ORP.
- ◆ Without knowing OPR we don't know how "effective" the disinfectant is.
- ◆ Chlorine PPM measurement method is old and inappropriate which only shows residue of Chlorine. ORP is latest and WHO recommended which shows active chlorine dioxide effective in water.

Competitive Benefits

- ◆ Only Patented and FDA approved product in the market which ensures product safety and quality consistency with
- ◆ Certifications like GMP, WHO GMP, ISO and HALAL
- ◆ Highest Purity of more than 99% for no other DBPs other than fractions of Sodium Sulphate & Sodium Chloride.
- ◆ Optimum Safe concentration of 7.5% & 0.75% with almost no explosion hazards for better economy.
- ◆ Proven for performance in more than 3000 poultry farms in India including Air Disinfectant.



Product Packing

Boom-Ox (ClO₂)- 0.75%

Stock Solution Method : Single Can of 2375 ml (Part A) + Small attached can of 125 ml (Part B), Mix Part A & Part B and keep this stock solution air tight bottle for 1 hr, then ready to use.

Dose : 50-200 ml/1000 Litre water (Dose depends on source water ORP)

Pack : 2375ml (Part A) + 125ml (Part B) = 2500 ml (Part A+ Part B = 2.5 litre single can)

Boom - Ox (ClO₂) - 7.5%

Stock Solution Method : 50ml part A + 50 ml part B mix into 900 ml potable water. After mixing keep for 1 hr. in air tight bottle and use this ready stock solution for dosing.

Dose : Dosage depends on source water ORP and water hold duration in shed.

Approx Dose : 50-200 ml /1000 Litre water

Pack : 5 Ltr & 25 Ltr

General Guidelines-

- 1) Always keep air tight stock solution and product bottle.
- 2) Stock solution and Product pack keep in dry and dry places for Part A and Part B.
- 3) Do not keep product and stock solution near electric, magnetic field, direct sunlight and hot condition (Part A + Part B together)
- 4) Once Boom-Ox added in water tank, keep tank closed air tight and cool, in shed to avoid direct sun light.
- 5) Add calculated dose of Boom-Ox once tank is full. Do not add Boom-Ox dose before filling water tank.
- 6) For any enquiry and feedback please contact to our company expert.

Manufactured by :



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